day_4_Debarshi

Debarshi Das Adhikary

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Define the directory path containing our CEL files

```
celpath <- "/home/ibab/bio-env"
```

Load packages for microarray processing

```
library(affy)
```

```
## Loading required package: BiocGenerics
```

```
## Loading required package: generics
```

```
##
## Attaching package: 'generics'
```

```
## The following objects are masked from 'package:base':
##
##     as.difftime, as.factor, as.ordered, intersect, is.element, setdiff,
##     setequal, union
```

```
##
## Attaching package: 'BiocGenerics'
```

```
## The following objects are masked from 'package:stats':
##
##     IQR, mad, sd, var, xtabs
```

```
## The following objects are masked from 'package:base':
##
##     anyDuplicated, aperm, append, as.data.frame, basename, cbind,
##     colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
##     get, grep, grepl, is.unsorted, lapply, Map, mapply, match, mget,
##     order, paste, pmax, pmax.int, pmin, pmin.int, Position, rank,
##     rbind, Reduce, rownames, sapply, saveRDS, table, tapply, unique,
##     unsplit, which.max, which.min
```

```
## Loading required package: Biobase
```

```
## Welcome to Bioconductor
##
##     Vignettes contain introductory material; view with
##     'browseVignettes()'. To cite Bioconductor, see
##     'citation("Biobase")', and for packages 'citation("pkgname")'.
```

```
library(limma)
```

```
##
## Attaching package: 'limma'
```

```
## The following object is masked from 'package:BiocGenerics':
##
##     plotMA
```

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.2.1     ✔ readr     2.2.0
## ✔ forcats   1.0.1     ✔ stringr   1.6.0
## ✔ ggplot2   4.0.3     ✔ tibble    3.3.1
## ✔ lubridate 1.9.5     ✔ tidyr     1.3.2
## ✔ purrr     1.2.2
```

```
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::combine()    masks Biobase::combine(), BiocGenerics::combine()
## ✖ dplyr::filter()     masks stats::filter()
## ✖ dplyr::lag()        masks stats::lag()
## ✖ lubridate::pm()     masks affy::pm()
## ✖ ggplot2::Position() masks BiocGenerics::Position(), base::Position()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Read raw microarray data from the specified directory

```
yeast_microchip <- ReadAffy(celfile.path = celpath, compress = TRUE)
```

Extract gene expression matrix into a standard data frame

```
df_yeast <- data.frame(exprs(yeast_microchip))
```

Preview the top rows of the expression matrix

```
head(df_yeast)
```

	EWE11465_0001_WT1.CEL	EWE11465_0002_WT2.CEL	EWE11465_0003_WT3.CEL	EWE11465_0004_drna1.CEL
	<dbl>	<dbl>	<dbl>	<dbl>
1	75	95	83	71
2	4530	4222	4416	4636
3	74	91	78	85
4	4918	4362	4714	4557
5	111	79	80	93
6	66	87	77	66

6 rows | 1-5 of 7 columns

Extract Perfect Match (PM) probe signals

```
pm_yeast <- data.frame(affy::pm(yeast_microchip))
```

```
## Warning: replacing previous import 'AnnotationDbi::tail' by 'utils::tail' when
## loading 'ecoli2cdf'
```

```
## Warning: replacing previous import 'AnnotationDbi::head' by 'utils::head' when
## loading 'ecoli2cdf'
```

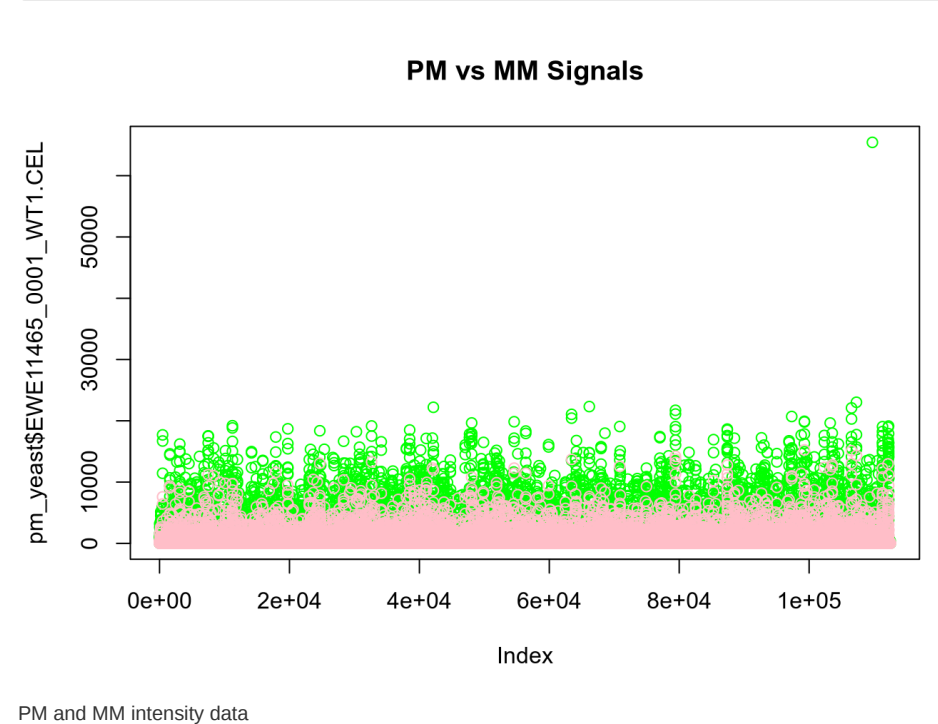
```
##
```

Extract Mismatch (MM) probe signals

```
mm_yeast <- data.frame(affy::mm(yeast_microchip))
```

Scatter plot of PM intensities for the first Wild Type sample

```
plot(pm_yeast$EWE11465_0001_WT1.CEL, col = "green", main = "PM vs MM Signals")
# Overlay MM intensities on the same plot
points(mm_yeast$EWE11465_0001_WT1.CEL, col = "pink")
```

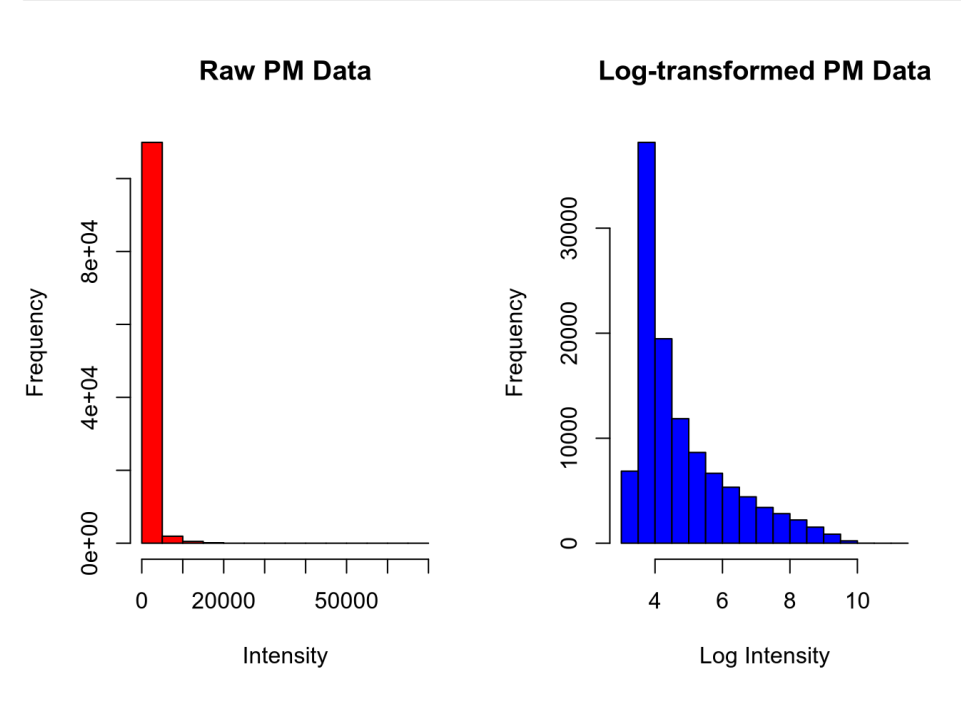


## Performed log-transformation on PM and MM intensity data

```
pmlog <- log(pm_yeast)
mmlog <- log(mm_yeast)
```

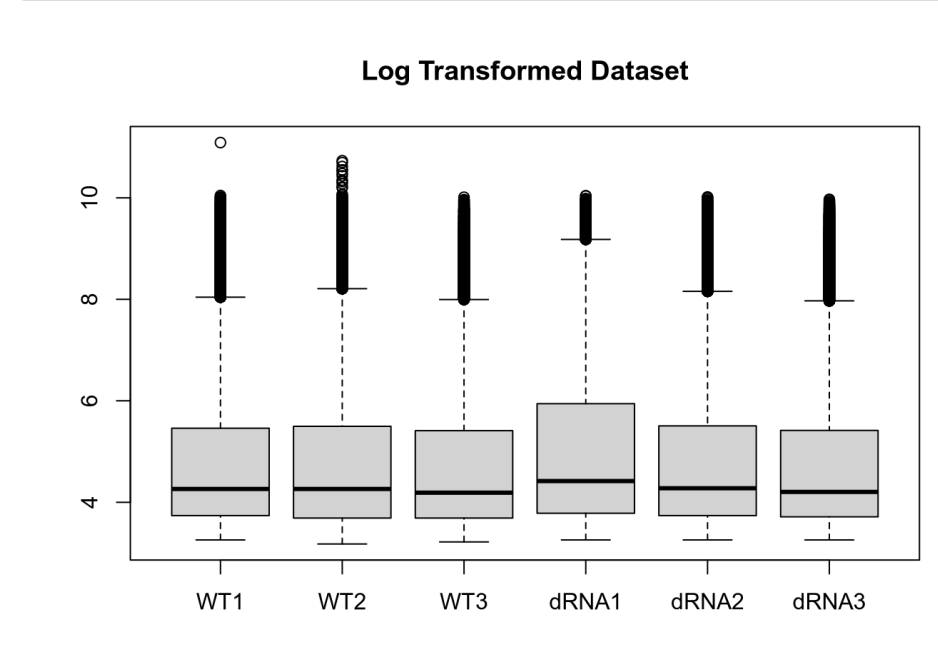
Set up layout for two side-by-side plots

```
par(mfrow = c(1, 2))

# Histogram of raw PM values
hist(pm_yeast$EWE11465_0001_WT1.CEL, main = "Raw PM Data", xlab = "Intensity", col = "red")

# Histogram of log-transformed PM values
hist(pmlog$EWE11465_0001_WT1.CEL, main = "Log-transformed PM Data", xlab = "Log Intensity", col = "blue")
```



Extract individual sample log-PM vectors

```
wt1_log  <- pmlog$EWE11465_0001_WT1.CEL
wt2_log  <- pmlog$EWE11465_0002_WT2.CEL
wt3_log  <- pmlog$EWE11465_0003_WT3.CEL
drna1_log <- pmlog$EWE11465_0004_drna1.CEL
drna2_log <- pmlog$EWE11465_0005_drna2.CEL
drna3_log <- pmlog$EWE11465_0006_drna3.CEL

# Plot boxplots across samples
boxplot(wt1_log, wt2_log, wt3_log, drna1_log, drna2_log, drna3_log,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),main = "Log Transformed Dataset")
```



##Applied RMA

```
yeast_microchip_rma <- rma(yeast_microchip)
```

```
## Background correcting
## Normalizing
## Calculating Expression
```

##Extraced gene expression matrix into a standard data frame

```
df_rma_yeast <- data.frame(exprs(yeast_microchip_rma))

#the top rows of the expression matrix
head(df_rma_yeast)
```

	EWE11465_0001_WT1.CEL	EWE11465_0002_WT2.CEL	EWE11465_0003_WT3.CEL
	<dbl>	<dbl>	<dbl>
1759068_at	3.413791	3.015579	3.437268
1759069_at	9.020832	9.126874	9.048895
1759070_s_at	7.771956	7.623827	7.432955
1759071_s_at	6.148551	5.695259	5.836691
1759072_s_at	3.670370	3.580104	3.662050
1759073_at	2.925485	3.253104	2.651791

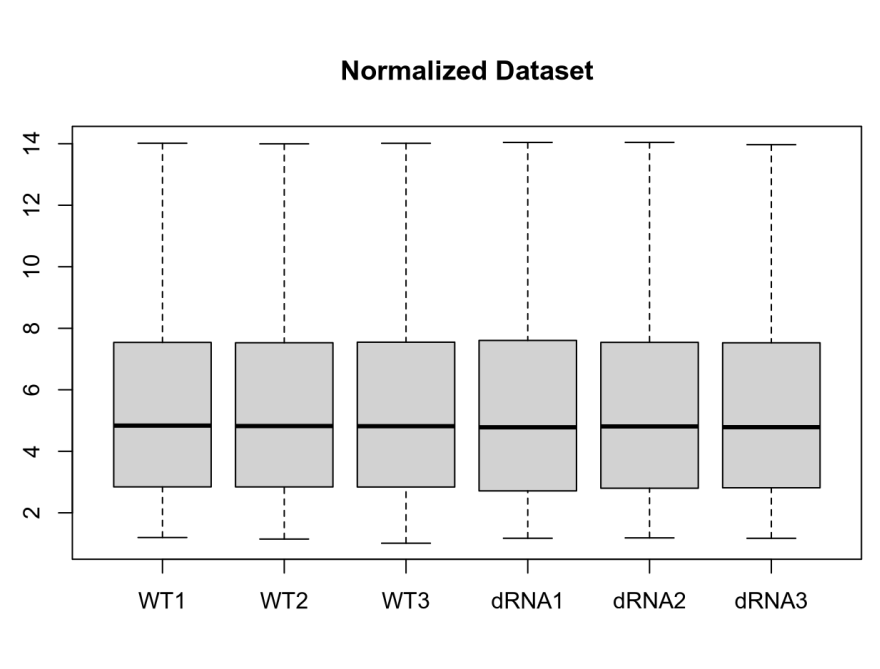
6 rows | 1-4 of 7 columns

##Stored individual sample vectors from the RMA-normalized data frame

```
wt1_norm  <- df_rma_yeast$EWE11465_0001_WT1.CEL
wt2_norm  <- df_rma_yeast$EWE11465_0002_WT2.CEL
wt3_norm  <- df_rma_yeast$EWE11465_0003_WT3.CEL
drna1_norm <- df_rma_yeast$EWE11465_0004_drna1.CEL
drna2_norm <- df_rma_yeast$EWE11465_0005_drna2.CEL
drna3_norm <- df_rma_yeast$EWE11465_0006_drna3.CEL
```

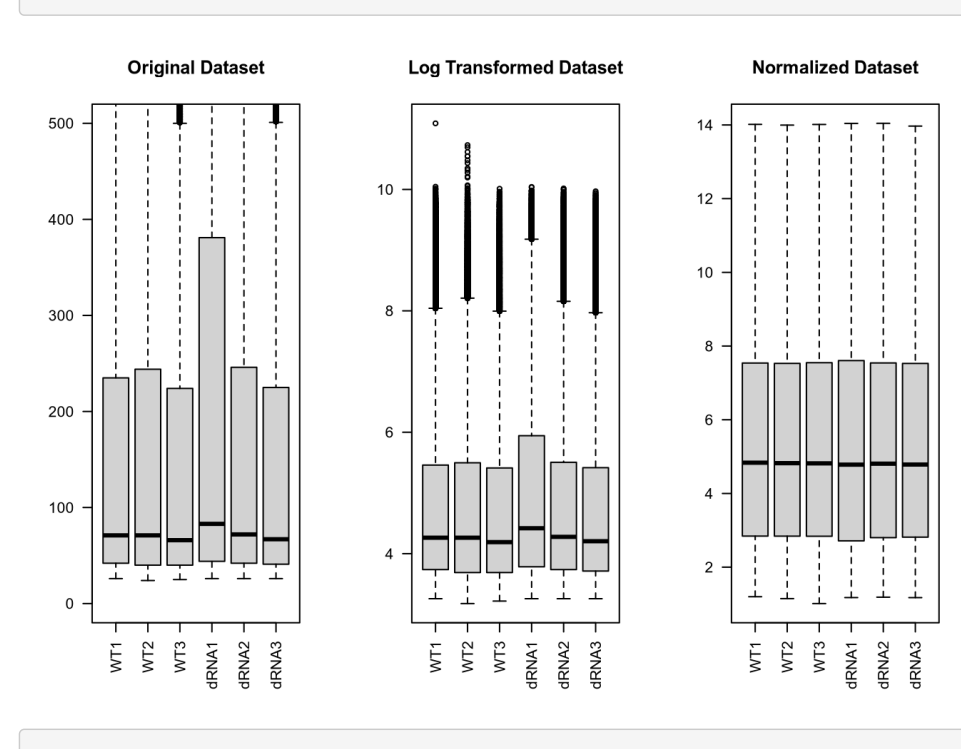
##Boxplot of RMA-normalized expression values

```
boxplot(wt1_norm, wt2_norm, wt3_norm, drna1_norm, drna2_norm, drna3_norm,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),main = "Normalized Dataset")
```



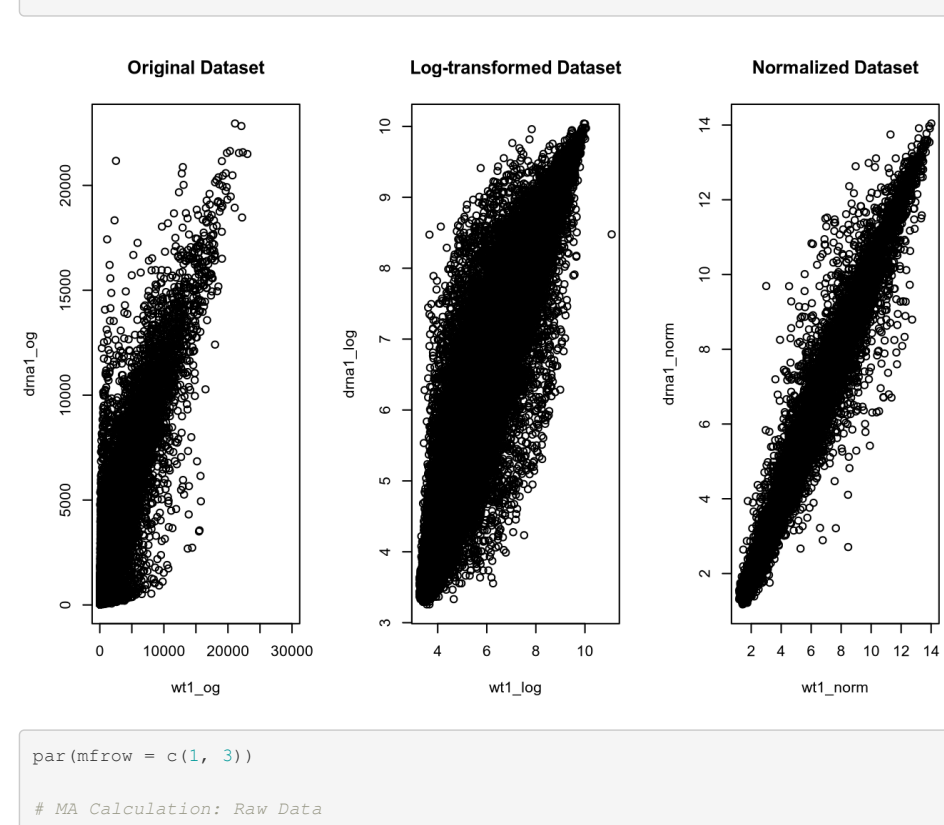
##Prepared raw vectors for comparison

```
wt1_og  <- pm_yeast$EWE11465_0001_WT1.CEL
wt2_og  <- pm_yeast$EWE11465_0002_WT2.CEL
wt3_og  <- pm_yeast$EWE11465_0003_WT3.CEL
drna1_og <- pm_yeast$EWE11465_0004_drna1.CEL
drna2_og <- pm_yeast$EWE11465_0005_drna2.CEL
drna3_og <- pm_yeast$EWE11465_0006_drna3.CEL

# Display three side-by-side boxplots
par(mfrow = c(1, 3))

# 1. Original raw dataset
boxplot(wt1_og, wt2_og, wt3_og, drna1_og, drna2_og, drna3_og,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),
        ylim = c(0, 500), main = "Original Dataset", las = 2)

# 2. Log-transformed dataset
boxplot(wt1_log, wt2_log, wt3_log, drna1_log, drna2_log, drna3_log,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),main = "Log Transformed Dataset", las = 2)

# 3. RMA-normalized dataset
boxplot(wt1_norm, wt2_norm, wt3_norm, drna1_norm, drna2_norm, drna3_norm,
        names = c("WT1", "WT2", "WT3", "dRNA1", "dRNA2", "dRNA3"),main = "Normalized Dataset", las = 2)
```



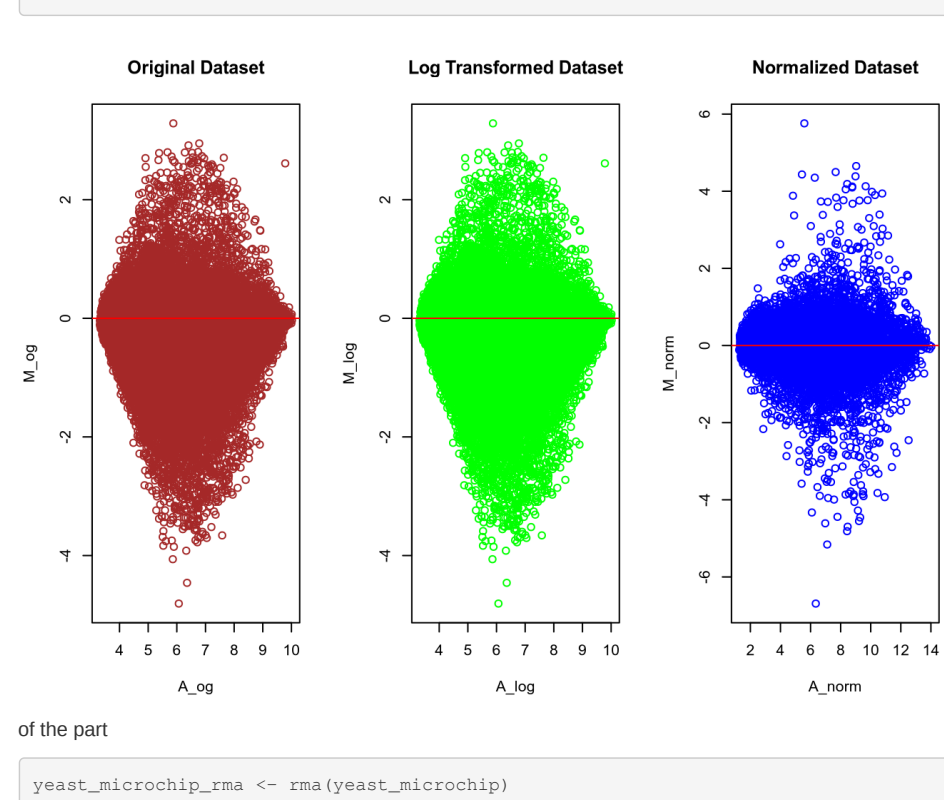
##compared scatter plots

```
par(mfrow = c(1, 3))

# Raw scatter plot
plot(wt1_og, drna1_og, main = "Original Dataset", xlim = c(0, 30000))

# Log-transformed scatter plot
plot(wt1_log, drna1_log, main = "Log-transformed Dataset")

# Normalized scatter plot
plot(wt1_norm, drna1_norm, main = "Normalized Dataset")
```



##Generated MA plots

```
par(mfrow = c(1, 3))

# MA Calculation: Raw Data
A_og <- (log(wt1_og) + log(drna1_og)) / 2
M_og <- log(wt1_og / drna1_og)
plot(A_og, M_og, main = "Original Dataset", col='brown')
abline(h = 0, col = "red")

# MA Calculation: Log-Transformed Data
A_log <- (wt1_log + drna1_log) / 2
M_log <- wt1_log - drna1_log
plot(A_log, M_log, main = "Log Transformed Dataset", col='green')
abline(h = 0, col = "red")

# MA Calculation: Normalized Data
A_norm <- (wt1_norm + drna1_norm) / 2
M_norm <- wt1_norm - drna1_norm
plot(A_norm, M_norm, main = "Normalized Dataset", col='blue')
abline(h = 0, col = "red")
```



##This will act as an input for the rest of the part

```
yeast_microchip_rma <- rma(yeast_microchip)
```

```
## Background correcting
## Normalizing
## Calculating Expression
```

```
library(limma)
```

##read the RMA normalized values , printing that we can get to know the exact order

```
pData_yest = pData(yeast_microchip_rma)
pData_yest
```

	sample
	<int>
EWE11465_0001_WT1.CEL	1
EWE11465_0002_WT2.CEL	2
EWE11465_0003_WT3.CEL	3
EWE11465_0004_drna1.CEL	4
EWE11465_0005_drna2.CEL	5
EWE11465_0006_drna3.CEL	6

6 rows

##We are grouping this,it groups first three as WT group and next three as Mutant, if it is in alternate order , we can write as WT, mutant, WT, mutant like this… ##Groups =pData_yeast$Treatment, order matters here, we can see two levels,if we write WT1 by mistake it will take that as another level, so instead of two here it will become three and WT1 will not have replicates.

```
Groups = c("WT","WT","WT","Mutant","Mutant","Mutant")
Groups
```

```
## [1] "WT"     "WT"     "WT"     "Mutant" "Mutant" "Mutant"
```

##Purpose: Convert the character vector into a factor ##factors take the parameters in a alphanumeric way,which automatically detects what to compare with whom,,,order does not matter here ##relevel tells what we are going to use as reference..it prevents use of that alphanumeric way,,,and specifically tells what to use as reference ##but use of factor is important to tell how many levels are here.

```
Groups = factor(Groups, levels = c("WT","Mutant"))
Groups = relevel(Groups, ref= "WT")
Groups
```

```
## [1] WT     WT     WT     Mutant Mutant Mutant
## Levels: WT Mutant
```

##We have to create a design matrix, here first three columns, it will take as wt because it has values of 1 for gene 1 and so on and will take the avg..

```
design = model.matrix(~0+Groups)

##design = model.matrix(~0+Groups)..the first 0 ...it  tells to take intercept as 0. No intercept model is also c
alled as means model..without 0,,it will keep the intercept and that will be the mean reference model..there I wi
ll get only one column that is mutant.
##IN the means model,,if i get 9 in WT and * in Mutant ,,that states the avg intensity,,but in mean reference mod
el the values are in ratio..

design
```

```
##   GroupsWT GroupsMutant
## 1        1            0
## 2        1            0
## 3        1            0
## 4        0            1
## 5        0            1
## 6        0            1
## attr(,"assign")
## [1] 1 1
## attr(,"contrasts")
## attr(,"contrasts")$Groups
## [1] "contr.treatment"
```

##Here the first three columns is 1 for WT,,,so while calculating the avg intensities from gene 1 for wild type it will take the first three column's intensities and put in the formula of B0WT+B1MT,,where it will multiply 0 with the values of next three columns which is effectively mutant types,,,so we will get avg intensity values for WT for gene1. but the avg will be calculated using lmfit…

```
fit_y = lmFit(yeast_microchip_rma, design)
contrast.matrix = makeContrasts(GroupsMutant-GroupsWT, levels = design)
##Considering WT as reference, and we are comparing against mutants. This is important, if the values are positiv
e it will say that the genes are upregulated in mutants.
data.fit.con = contrasts.fit(fit_y,contrast.matrix)
##contrast matrix and design matrix does not  have the avg intensity values, so we are using fit_y here,,fit_y ha
ve the avg intensity values, variances , SE etc of all the columns , defined by the design matrix. And if thre ar
e multiple groups like WT1, MT1, MT2--to compare and specify which two we are going to compare we can use contras
t matrix.
fit12 = eBayes(data.fit.con)
##Here by eBayes we are using the moderated t-test. It borrows variance info from all the other genes and then ca
lulate the avg and adjust the  variances according to that.
contrast.matrix
```

```
##               Contrasts
## Levels         GroupsMutant - GroupsWT
##   GroupsWT                          -1
##   GroupsMutant                       1
```

```
fit_y
```

```
## An object of class "MArrayLM"
## $coefficients
##              GroupsWT GroupsMutant
## 1759068_at   3.288879     3.046389
## 1759069_at   9.065534     8.543857
## 1759070_s_at 7.609579     8.908647
## 1759071_s_at 5.893500     5.619717
## 1759072_s_at 3.637508     3.588137
## 10203 more rows ...
##
## $rank
## [1] 2
##
## $assign
## [1] 1 1
##
## $qr
## $qr
##     GroupsWT GroupsMutant
## 1 -1.7320508    0.0000000
## 2  0.5773503   -1.7320508
## 3  0.5773503    0.0000000
## 4  0.0000000    0.5773503
## 5  0.0000000    0.5773503
## 6  0.0000000    0.5773503
## attr(,"assign")
## [1] 1 1
## attr(,"contrasts")
## attr(,"contrasts")$Groups
## [1] "contr.treatment"
##
##
## $qraux
## [1] 1.57735 1.00000
##
## $pivot
## [1] 1 2
##
## $tol
## [1] 1e-07
##
## $rank
## [1] 2
##
##
## $df.residual
## [1] 4 4 4 4 4
## 10203 more elements ...
##
## $sigma
##   1759068_at   1759069_at 1759070_s_at 1759071_s_at 1759072_s_at
##    0.2023033    0.1899167    0.2355675    0.4926051    0.2171652
## 10203 more elements ...
##
## $cov.coefficients
##               GroupsWT GroupsMutant
## GroupsWT     0.3333333    0.0000000
## GroupsMutant 0.0000000    0.3333333
##
## $stdev.unscaled
##               GroupsWT GroupsMutant
## 1759068_at   0.5773503    0.5773503
## 1759069_at   0.5773503    0.5773503
## 1759070_s_at 0.5773503    0.5773503
## 1759071_s_at 0.5773503    0.5773503
## 1759072_s_at 0.5773503    0.5773503
## 10203 more rows ...
##
## $pivot
## [1] 1 2
##
## $Amean
##   1759068_at   1759069_at 1759070_s_at 1759071_s_at 1759072_s_at
##     3.167634     8.804695     8.259113     5.756609     3.612822
## 10203 more elements ...
##
## $method
## [1] "ls"
##
## $design
##   GroupsWT GroupsMutant
## 1        1            0
## 2        1            0
## 3        1            0
## 4        0            1
## 5        0            1
## 6        0            1
## attr(,"assign")
## [1] 1 1
## attr(,"contrasts")
## attr(,"contrasts")$Groups
## [1] "contr.treatment"
```

##makeContrasts is for deifing contrasts, where it is telling to take the differences between mutant and wild type. So it is like specifically defining condition of interest for us.

```
fit12
```

```
## An object of class "MArrayLM"
## $coefficients
##               Contrasts
##                GroupsMutant - GroupsWT
##   1759068_at               -0.24249026
##   1759069_at               -0.52167654
##   1759070_s_at              1.29906777
##   1759071_s_at             -0.27378326
##   1759072_s_at             -0.04937133
## 10203 more rows ...
##
## $rank
## [1] 2
##
## $assign
## [1] 1 1
##
## $qr
## $qr
##     GroupsWT GroupsMutant
## 1 -1.7320508    0.0000000
## 2  0.5773503   -1.7320508
## 3  0.5773503    0.0000000
## 4  0.0000000    0.5773503
## 5  0.0000000    0.5773503
## 6  0.0000000    0.5773503
## attr(,"assign")
## [1] 1 1
## attr(,"contrasts")
## attr(,"contrasts")$Groups
## [1] "contr.treatment"
##
##
## $qraux
## [1] 1.57735 1.00000
##
## $pivot
## [1] 1 2
##
## $tol
## [1] 1e-07
##
## $rank
## [1] 2
##
##
## $df.residual
## [1] 4 4 4 4 4
## 10203 more elements ...
##
## $sigma
##   1759068_at   1759069_at 1759070_s_at 1759071_s_at 1759072_s_at
##    0.2023033    0.1899167    0.2355675    0.4926051    0.2171652
## 10203 more elements ...
##
## $cov.coefficients
##                          Contrasts
## Contrasts                 GroupsMutant - GroupsWT
##   GroupsMutant - GroupsWT               0.6666667
##
## $stdev.unscaled
##               Contrasts
##                GroupsMutant - GroupsWT
##   1759068_at                 0.8164966
##   1759069_at                 0.8164966
##   1759070_s_at               0.8164966
##   1759071_s_at               0.8164966
##   1759072_s_at               0.8164966
## 10203 more rows ...
##
## $pivot
## [1] 1 2
##
## $Amean
##   1759068_at   1759069_at 1759070_s_at 1759071_s_at 1759072_s_at
##     3.167634     8.804695     8.259113     5.756609     3.612822
## 10203 more elements ...
##
## $method
## [1] "ls"
##
## $design
##   GroupsWT GroupsMutant
## 1        1            0
## 2        1            0
## 3        1            0
## 4        0            1
## 5        0            1
## 6        0            1
## attr(,"assign")
## [1] 1 1
## attr(,"contrasts")
## attr(,"contrasts")$Groups
## [1] "contr.treatment"
##
##
## $contrasts
##               Contrasts
## Levels         GroupsMutant - GroupsWT
##   GroupsWT                          -1
##   GroupsMutant                       1
##
## $df.prior
## [1] 2.791656
##
## $s2.prior
## [1] 0.03315848
##
## $var.prior
## [1] 112.7717
##
## $proportion
## [1] 0.01
##
## $s2.post
##   1759068_at   1759069_at 1759070_s_at 1759071_s_at 1759072_s_at
##   0.03773360   0.03487228   0.04631200   0.15654593   0.04140522
## 10203 more elements ...
##
## $t
##               Contrasts
##                GroupsMutant - GroupsWT
##   1759068_at                -1.5288886
##   1759069_at                -3.4214229
##   1759070_s_at               7.3931697
##   1759071_s_at              -0.8474842
##   1759072_s_at              -0.2971617
## 10203 more rows ...
##
## $df.total
## [1] 6.791656 6.791656 6.791656 6.791656 6.791656
## 10203 more elements ...
##
## $p.value
##               Contrasts
##                GroupsMutant - GroupsWT
##   1759068_at              0.1714374367
##   1759069_at              0.0116438151
##   1759070_s_at            0.0001742597
##   1759071_s_at            0.4255790053
##   1759072_s_at            0.7752253329
## 10203 more rows ...
##
## $lods
##               Contrasts
##                GroupsMutant - GroupsWT
##   1759068_at                 -6.019052
##   1759069_at                 -3.299296
##   1759070_s_at                1.237194
##   1759071_s_at               -6.774273
##   1759072_s_at               -7.113453
## 10203 more rows ...
##
## $F
## [1]  2.3375003 11.7061348 54.6589575  0.7182295  0.0883051
## 10203 more elements ...
##
## $F.p.value
## [1] 0.1714374367 0.0116438151 0.0001742597 0.4255790053 0.7752253329
## 10203 more elements ...
```

##from thr eBayes we got log of Odds, pvalue etc…top table is used

```
diffexp= topTable(fit12, number=Inf,adjust.method="BH")
# adjusted p value calulated basd on Benjamin hotchback correction and we are writing Inf to calulate fold diff a
nd adj p value for all the genes/probes. It  prints in the order of higest expression difference to lowest. If I
have given 10 in the "number"..it will calculate for only 10 highest,,,
diffexp
```

	logFC	AveExpr	t	P.Value	adj.P.Val	B
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1766647_s_at	-5.933308354	5.554242	-54.19340480	3.280059e-10	3.348284e-06	11.564631769
AFFX-Ec-U57609-2_s_at	7.155443921	6.466314	39.60308717	2.744066e-09	1.400571e-05	10.714404789
1769071_s_at	-2.820357603	10.795486	-25.62751217	5.180489e-08	1.241138e-04	8.933794142
1765239_s_at	-2.983623681	9.268889	-25.25463886	5.717466e-08	1.241138e-04	8.861777685
1766702_s_at	4.506791324	9.075793	24.20856534	7.599151e-08	1.241138e-04	8.649837382
1764012_s_at	2.838458037	8.984151	23.81989936	8.472505e-08	1.241138e-04	8.567175727
1762388_s_at	3.329536941	5.449938	23.40879016	9.523913e-08	1.241138e-04	8.477307029
1763982_s_at	5.848421046	7.352742	23.18751660	1.015131e-07	1.241138e-04	8.427869607
1766364_s_at	2.570209931	8.176837	22.67392865	1.179888e-07	1.241138e-04	8.310146743
1762983_s_at	3.976983275	10.713461	22.57274918	1.215848e-07	1.241138e-04	8.286452033

1-10 of 10,000 rows Previous 1 2 3 4 5 6 ... 1000 Next

```
diffexp = topTable(fit12,
number = Inf,
adjust.method = "BH")
diffexp
```

	logFC	AveExpr	t	P.Value	adj.P.Val	B
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1766647_s_at	-5.933308354	5.554242	-54.19340480	3.280059e-10	3.348284e-06	11.564631769
AFFX-Ec-U57609-2_s_at	7.155443921	6.466314	39.60308717	2.744066e-09	1.400571e-05	10.714404789
1769071_s_at	-2.820357603	10.795486	-25.62751217	5.180489e-08	1.241138e-04	8.933794142
1765239_s_at	-2.983623681	9.268889	-25.25463886	5.717466e-08	1.241138e-04	8.861777685
1766702_s_at	4.506791324	9.075793	24.20856534	7.599151e-08	1.241138e-04	8.649837382
1764012_s_at	2.838458037	8.984151	23.81989936	8.472505e-08	1.241138e-04	8.567175727
1762388_s_at	3.329536941	5.449938	23.40879016	9.523913e-08	1.241138e-04	8.477307029
1763982_s_at	5.848421046	7.352742	23.18751660	1.015131e-07	1.241138e-04	8.427869607
1766364_s_at	2.570209931	8.176837	22.67392865	1.179888e-07	1.241138e-04	8.310146743
1762983_s_at	3.976983275	10.713461	22.57274918	1.215848e-07	1.241138e-04	8.286452033

1-10 of 10,000 rows Previous 1 2 3 4 5 6 ... 1000 Next

```
#the fold change is in log2 scale, 1 means 2 fold upreguation and -1 means 2 fold downregualtion ,,,,0.6 means 1.
5 fold upregualtion.
```

now plot a volcano plot between log10(Adjpvalue) and logFC…to get to now that pvalue cut off suits the most for our data.

```
# Upregulated genes
up = subset(diffexp,
logFC >= 0.6 &
adj.P.Val <= 0.05)
up
```

	logFC	AveExpr	t	P.Value	adj.P.Val	B
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
AFFX-Ec-U57609-2_s_at	7.1554439	6.466314	39.603087	2.744066e-09	1.400571e-05	10.714404789
1766702_s_at	4.5067913	9.075793	24.208565	7.599151e-08	1.241138e-04	8.649837382
1764012_s_at	2.8384580	8.984151	23.819899	8.472505e-08	1.241138e-04	8.567175727
1762388_s_at	3.3295369	5.449938	23.408790	9.523913e-08	1.241138e-04	8.477307029
1763982_s_at	5.8484210	7.352742	23.187517	1.015131e-07	1.241138e-04	8.427869607
1766364_s_at	2.5702099	8.176837	22.673929	1.179888e-07	1.241138e-04	8.310146743
1762983_s_at	3.9769833	10.713461	22.572749	1.215848e-07	1.241138e-04	8.286452033
1767171_s_at	5.4165354	8.731017	21.104819	1.908717e-07	1.603383e-04	7.922863245
1761245_s_at	4.7062248	9.451509	20.748984	2.139033e-07	1.603383e-04	7.828804986
1768555_s_at	3.9682202	9.135292	20.655579	2.204659e-07	1.603383e-04	7.803709223

1-10 of 636 rows Previous 1 2 3 4 5 6 ... 64 Next

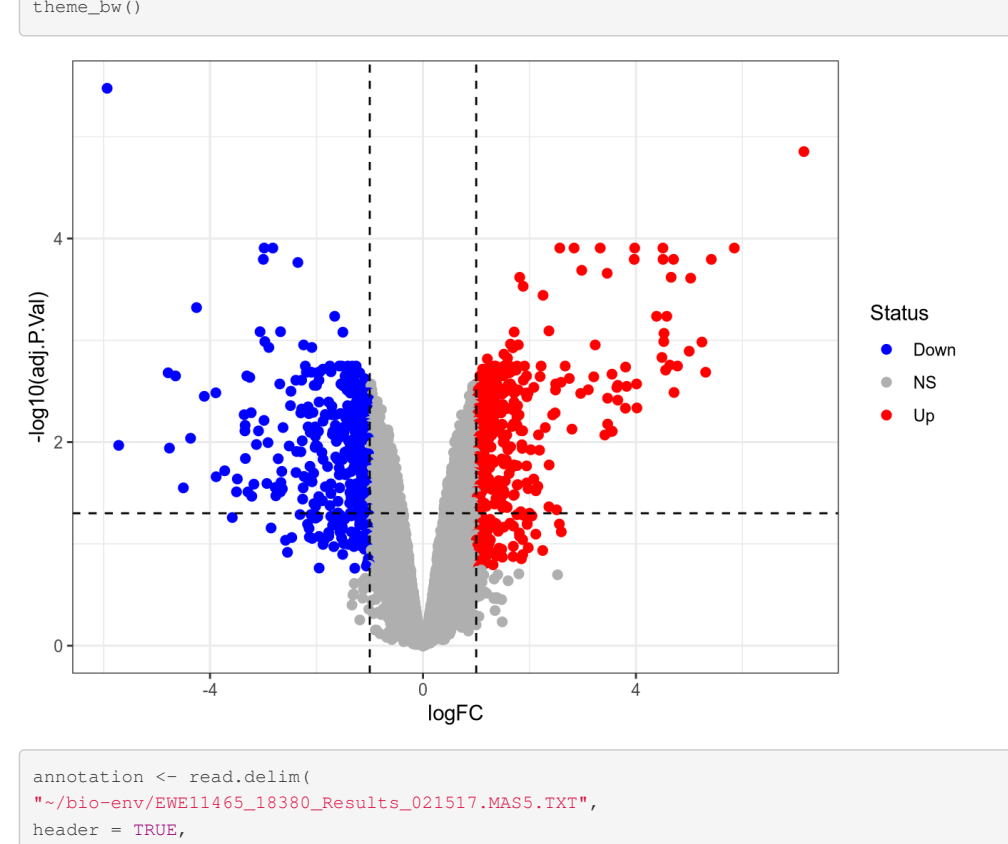
we dont have gene id we have prob id,,there is a annotation file we have downloded, there it is present.

```
# Downregulated genes
down = subset(diffexp,
logFC <= -0.6 &
adj.P.Val <= 0.05)
down
```

	logFC	AveExpr	t	P.Value	adj.P.Val	B
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1766647_s_at	-5.9333084	5.554242	-54.193405	3.280059e-10	3.348284e-06	11.56463177
1769071_s_at	-2.8203576	10.795486	-25.627512	5.180489e-08	1.241138e-04	8.93379414
1765239_s_at	-2.9836237	9.268889	-25.254639	5.717466e-08	1.241138e-04	8.86177769
1764675_at	-2.9989185	10.578554	-20.451710	2.356068e-07	1.603383e-04	7.74833559
1768362_s_at	-2.3504126	7.155229	-20.042286	2.697569e-07	1.721049e-04	7.63459494
1762126_s_at	-4.2537896	5.517309	-16.183494	1.122521e-06	4.774457e-04	6.36914149
1769163_s_at	-1.6587005	10.143291	-15.434414	1.537089e-06	5.811337e-04	6.07529688
1760483_at	-3.0603255	7.108772	-14.443420	2.383966e-06	8.236275e-04	5.65718101
1765917_s_at	-2.6757767	8.286600	-14.410150	2.420535e-06	8.236275e-04	5.64252251
1759328_s_at	-1.5048273	8.634835	-14.246087	2.610516e-06	8.327546e-04	5.56961951

1-10 of 554 rows Previous 1 2 3 4 5 6 ... 56 Next

```
diffexp$Status <- "NS"
diffexp$Status[
  diffexp$logFC >= 1 & diffexp$P.Value < 0.05
] <- "Up"
diffexp$Status[
diffexp$logFC <= -1 & diffexp$P.Value < 0.05
] <- "Down"
ggplot(diffexp,
aes(logFC, -log10(adj.P.Val), color = Status)) +
geom_point(size = 2) +
scale_color_manual(values = c(
"Up" = "red",
"Down" = "blue",
"NS" = "grey70"
)) +
geom_vline(xintercept = c(-1, 1), linetype = 2) +
geom_hline(yintercept = -log10(0.05), linetype = 2) +
theme_bw()
```



```
annotation <- read.delim(
"~/bio-env/EWE11465_18380_Results_021517.MAS5.TXT",
header = TRUE,
stringsAsFactors = FALSE,
check.names = FALSE
)
colnames(annotation)
```

```
##  [1] "Probe Set ID"                     "EWE11465-0001_WT1.mas5-Signal"
##  [3] "EWE11465-0002_WT2.mas5-Signal"    "EWE11465-0003_WT3.mas5-Signal"
##  [5] "EWE11465-0004_drna1.mas5-Signal"  "EWE11465-0005_drna2.mas5-Signal"
##  [7] "EWE11465-0006_drna3.mas5-Signal"  "Representative Public ID"
##  [9] "Gene Title"                       "Gene Symbol"
## [11] "Gene Ontology Biological Process" "Gene Ontology Cellular Component"
## [13] "Gene Ontology Molecular Function" "Pathway"
```

```
annotation_small <- annotation[, c("Probe Set ID",
"Representative Public ID",
"Gene Symbol",
"Gene Title")]
diffexp$`Probe Set ID` <- rownames(diffexp)
annotation_small
```

Probe Set ID	Representative Public ID	Gene Symbol
<chr>	<chr>	<chr>
1759068_at	CFT073_c4728	---
1759069_at	MG1655_mrr_b4351	mrr
1759070_s_at	IG_1349064_1349430-r	---
1759071_s_at	MG1655_b2503	ECs3365 /// yfgF
1759072_s_at	EDL933_Z0312	ECs1611 /// ECs2987
1759073_at	MG1655_ycfK_b1154	stfP
1759074_at	MG1655_yfbN_b2273	yfbN
1759075_at	MG1655_ggt_b3447	ggt
1759076_s_at	IG_320306_320831-r	---
1759077_s_at	EDL933_Z1340	ECs1076

1-10 of 10,000 rows | 1-3 of 4 columns Previous 1 2 3 4 5 6 ... 1000 Next

```
library(dplyr)
diffexp_annotated <- left_join(diffexp,
annotation_small,
by = "Probe Set ID")
diffexp_annotated
```

logFC	AveExpr	t	P.Value	adj.P.Val	B	Status
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
-5.933308354	5.554242	-54.19340480	3.280059e-10	3.348284e-06	11.564631769	Down
7.155443921	6.466314	39.60308717	2.744066e-09	1.400571e-05	10.714404789	Up
-2.820357603	10.795486	-25.62751217	5.180489e-08	1.241138e-04	8.933794142	Down
-2.983623681	9.268889	-25.25463886	5.717466e-08	1.241138e-04	8.861777685	Down
4.506791324	9.075793	24.20856534	7.599151e-08	1.241138e-04	8.649837382	Up
2.838458037	8.984151	23.81989936	8.472505e-08	1.241138e-04	8.567175727	Up
3.329536941	5.449938	23.40879016	9.523913e-08	1.241138e-04	8.477307029	Up
5.848421046	7.352742	23.18751660	1.015131e-07	1.241138e-04	8.427869607	Up
2.570209931	8.176837	22.67392865	1.179888e-07	1.241138e-04	8.310146743	Up
3.976983275	10.713461	22.57274918	1.215848e-07	1.241138e-04	8.286452033	Up

1-10 of 10,000 rows | 1-7 of 11 columns Previous 1 2 3 4 5 6 ... 1000 Next

```
write.csv(diffexp_annotated,
"diffexp_annotated.csv",
row.names = FALSE)
```

```
up_annotated <- subset(diffexp_annotated, logFC >= 1)
up_annotated
```

	logFC	AveExpr	t	P.Value	adj.P.Val	B	Status
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>
2	7.155444	6.466314	39.603087	2.744066e-09	1.400571e-05	10.71440479	Up
5	4.506791	9.075793	24.208565	7.599151e-08	1.241138e-04	8.64983738	Up
6	2.838458	8.984151	23.819899	8.472505e-08	1.241138e-04	8.56717573	Up
7	3.329537	5.449938	23.408790	9.523913e-08	1.241138e-04	8.47730703	Up
8	5.848421	7.352742	23.187517	1.015131e-07	1.241138e-04	8.42786961	Up
9	2.570210	8.176837	22.673929	1.179888e-07	1.241138e-04	8.31014674	Up
10	3.976983	10.713461	22.572749	1.215848e-07	1.241138e-04	8.28645203	Up
11	5.416535	8.731017	21.104819	1.908717e-07	1.603383e-04	7.92286325	Up
12	4.706225	9.451509	20.748984	2.139033e-07	1.603383e-04	7.82880499	Up
13	3.968220	9.135292	20.655579	2.204659e-07	1.603383e-04	7.80370922	Up

1-10 of 438 rows | 1-8 of 12 columns Previous 1 2 3 4 5 6 ... 44 Next

```
down_annotated <- subset(diffexp_annotated, logFC <= -1)
```

```
write.csv(up_annotated, "Upregulated_Genes.csv", row.names = FALSE)
```

```
write.csv(down_annotated, "Downregulated_Genes.csv", row.names = FALSE)
```